

RKDF University, Bhopal
Faculty Profile



Basic Information				
Name	Dr. Ravi Kumar Singh Pippal			
Date of Birth	23 rd August 1983			
Designation	Professor and Dean			
Department	Computer Science and Engineering			
Experience	16 years			
Email ID	ravesingh@gmail.com			
Contact No	+91-9522708498			
Educational Qualifications				
Description	Year	%	Institute/University	
(UG)- B.E.	2005	63.19	Rajiv Gandhi Technological University, Bhopal (J.E.C. Jabalpur)	
(PG)- M. Tech. (Computer Science and Engineering)	2008	71.2	Rajiv Gandhi Technological University, Bhopal (S.A.T.I. Vidisha)	
M. Phil.	No	-	-	
(Ph. D.)- Computer Science and Engineering	2013	Awarded	ABV-Indian Institute of Information Technology and Management, Gwalior	
Post Doctorate	2022	Awarded	RKDF University, Bhopal	
NET Qualified/GATE	2009	91		
Experience Detail				
Experience (Teaching/Research)	Designation	Duration		Name of Institute/University
		From	To	
Teaching and Research	Professor	1 st Jan 2022	28 Oct. 2023	Vedica Institute of Technology, RKDF University, Bhopal (M.P.)
Teaching and Research	Associate Professor	1 st March 2016	31 Dec. 2021	Vedica Institute of Technology, RKDF University, Bhopal (M.P.)
Teaching	Assistant	6 th Feb. 2013	29 th Feb.	Radharaman Group of Institutes

	Professor		2016	(RITS, REC, RIRT), Bhopal (M.P.)
Teaching	Lecturer	16 th July 2008	16 th July 2009.	NRI Institute of Technology and Management, Gwalior (M.P.)
Publications				
No. of Papers Published (Attach Annexure of papers published in peer-reviewed or refereed journals)		97		
No. of Books Published		02		
Books Chapters Published (Provide print of 1st Page of Book		03		
No. of Patents Published/Grant		09		
Ph. D/M. Phil Project supervised				
Research Program		Award	Under Supervision	Name of University
Ph. D (Provide detail i.e. name, title etc)		09	-	RKDF University, Bhopal (M.P.)
M. Phil		-	-	-
PG Thesis/Dissertation		20	-	RKDF University, Bhopal (M.P.)
Area of Expertise (100 words)				
1) Symmetric Key and Public Key Cryptography 2) Steganography 3) Key Management 4) Image Encryption 5) Fuzzy Logic				
Award and Achievement				
Name of Award		Description (With certified of Copy award)		
National		-		
International		-		
Conference/Seminar/Workshops/FDP				
Description		No.		
Conference/Seminar p a p e r presentation		10		
Conference/Seminar attended/ organized		06		
Work shop attended/ organized		08		

FDP Attended/ organized	02	
Research Project		
Name of Project	Funding Agencies	Amount
Nil	-	-

• **Any other Achievement**

- 1) Program Co-ordinator for conducting Faculty Development Program on “Recent Trends in Computer Science and its Applications” under ISTE Institute Chapter.
- 2) Review research papers for many International Journals, out of which few are as follows.
 - **International Journal of Information Security (IJIS), Springer**
 - **Wireless Personal Communications (WPC), Springer**
 - **International Journal of Communication Systems (IJCS), Wiley**
 - **Security and Communication Networks (SCN), Wiley**
 - **Informatica**
 - **International Journal of Network Security (IJNS) and many more.**
- 3) No. of Conference Proceedings indexed in **Web of Science (THOMSON REUTERS Conference Proceedings Citation Index) – 04 (ICIEA’12, DBSec’12, ICCTD’10, WPMC’07)**
- 4) Technical Committee Member of **2nd International Conference on Advanced Computing and Intelligent Engineering (ICACIE 2017)** organized by Central University of Rajasthan, Ajmer, India, November 23-25, 2017.
- 5) Associate Editor-in-Chief for **SHODH SANGAM - A RKDF University Journal of Science and Engineering**, ISSN No.: 2581-5806.
- 6) Editor-in-Chief for **Research Journal of Engineering Technology and Management (RJETM)**, ISSN No.: 2582-6212.
- 7) Member of **International Journal of Computer Sciences and Engineering (IJCSE)**, ISSN No.: 2347-2693.
- 8) In the Editorial Board of **International Journal of Research in Technology (IJRT)**, ISSN No.: 2394-9007
- 9) In the Editorial Board of **International Journal of Scientific Research in Network Security and Communication (IJSRNSC)**, ISSN No.: 2321-3256
- 10) In the Reviewer Board of **International Journal of Computer Science & Engineering Technology (IJCSET)**, ISSN No.: 2231-0711

List of Patent Published

- 1) Patent application numbered **201721024152** dated **10-07-2017**
Title “An Apparatus and method based on use of Linear Interpolation for automated Drip Irrigation system in Agriculture using Wireless Sensor Network”. **[Published]**
- 2) Patent application numbered **202141037966** dated **22-08-2021**
Title “Touch Screen Devices Employing Graphene Electrodes”. **[Published]**
- 3) Patent application numbered **202141039102** dated **29-08-2021**
Title “Long Range Spy Robot with night vision”. **[Published]**
- 4) Patent application numbered **202221000273** dated **03-01-2022**
Title “Method for Migration of Sessions among Cipher Machines using Lagrange Interpolation”. **[Published]**
- 5) Patent application numbered **202121048051** dated **19-10-2021**
Title “Intelligent Water management process in crop cultivation using IOT devices”. **[Published]**
- 6) Patent application numbered **202121061606** dated **29-12-2021**
Title “An Intelligent resource management techniques for use at home-an eco-friendly approach”. **[Published]**
- 7) Patent application numbered **202221007249** dated **11-02-2022**
Title “Mechanism for Secure Key Establishment and Handover Management in Wireless Sensor Network”. **[Published]**
- 8) Patent application numbered **202221007499** dated **12-02-2022**
Title “Method and System for Secure Key Exchange in Internet Protocol Television (IPTV) Network”. **[Published]**
- 9) Patent application numbered **202221022518** dated **16-04-2022**
Title “Method for Thermal Energy Optimization for SPCC Model Using Evolutionary Algorithm”. **[Published]**

List of Books/Book Chapter published

- 1) Patel, J., Pippal, R.K.S. (2022). Graph-Based Mechanism to Prevent Structural Attack Over Social Media. In: Sharma, S., Peng, S.L., Agrawal, J., Shukla, R.K., Le, D.N. (eds) Data, Engineering and Applications. Lecture Notes in Electrical Engineering, vol 907. Springer, Singapore. https://doi.org/10.1007/978-981-19-4687-5_50 **[Indexed in SCOPUS]**
- 2) Rathore, R., Pippal, R.K.S. (2022). Semantic Relation-Based Modularity-Optimized Community Detection Algorithm for Heterogeneous Networks. In: Sharma, S., Peng, S.L., Agrawal, J., Shukla, R.K., Le, D.N. (eds) Data, Engineering and Applications. Lecture Notes in Electrical Engineering, vol 907. Springer, Singapore. https://doi.org/10.1007/978-981-19-4687-5_44 **[Indexed in SCOPUS]**
- 3) Kasbe T., Pippal R.S. (2019) Design and Implementation of Fuzzy Expert System for Dengue Diagnosis. In: Kamal R., Henshaw M., Nair P. (eds) International Conference on Advanced Computing Networking and Informatics. Advances in Intelligent Systems and Computing, vol 870. Springer, Singapore, ISBN: 978-981-13-2672-1. **[Indexed in SCOPUS]**

- 4) Pippal R.S., C.D. J., Tapaswi S. (2012) Security Vulnerabilities of User Authentication Scheme Using Smart Card. In: Cuppens-Boulahia N., Cuppens F., Garcia-Alfaro J. (eds) Data and Applications Security and Privacy XXVI. DBSec 2012. Lecture Notes in Computer Science, vol 7371. Springer, Berlin, Heidelberg, ISBN: 978-3-642-31539-8. **[Indexed in SCOPUS]**
- 5) Tanmay Kasbe and Ravi Singh Pippal, “An Object Oriented Programming with C++,” New Delhi, India, Idea Publishers, 2017, ISBN: 9788193270646.
- 6) Tanmay Kasbe and Ravi Singh Pippal, “Information Technology & MIS,” New Delhi, India, Idea Publishers, 2018, ISBN: 9788193326954.

List of Publications

- 1) Jitendra Patel, Ravi Kumar Singh Pippal, "An Approach to Prevent Neighborhood Attack over Social Media", Accepted for publication in *Samriddhi - A Journal of Physical Sciences, Engineering and Technology (S-JPSET)*, 2022. **[UGC Care Listed]**
- 2) Priyank Nayak, Ravi Kumar Singh Pippal, "Session Handover among Fog Devices using Polynomials", *Samriddhi - A Journal of Physical Sciences, Engineering and Technology (S-JPSET)*, vol. 14, no. 3, 2022, pp. 376-380. **[UGC Care Listed]**
- 3) Rishank Rathod, Ravi Kumar Singh Pippal, "Multi-Purpose function based Parliamentary Optimization Framework for Community Detection over social media", *Samriddhi - A Journal of Physical Sciences, Engineering and Technology (S-JPSET)*, vol. 14, no. 3, 2022, pp. 307-322. **[UGC Care Listed]**
- 4) Jitendra Patel, Ravi Kumar Singh Pippal, "Graph-based Mechanism to Prevent Structural Attack over Social Media", *Samriddhi - A Journal of Physical Sciences, Engineering and Technology (S-JPSET)*, vol. 14, no. 3, 2022, pp. 227-233. **[UGC Care Listed]**
- 5) Rishank Rathore, Ravi Kumar Singh Pippal, "Comprehensive Analysis of Single and Multi-Purpose Function-Based Community Detection over Social Media," *International Journal of Engineering Trends and Technology*, vol. 70, no. 2, 2022, pp. 86-102. **[Indexed in SCOPUS]**
- 6) Ravi Kumar Singh Pippal, Vinod Krishna Sethi, "Various Technology options of 'Application' of CCS on large coal-fired unit – A Rationale," *Samriddhi - A Journal of Physical Sciences, Engineering and Technology (S-JPSET)*, vol. 14, no. 1, 2022, pp. 55-60. **[UGC Care Listed]**
- 7) Girdhar Gopal Ladha, Ravi Kumar Singh Pippal, "A Computation Analysis to Predict Diabetes based on Data Mining: Detail Study," *Samriddhi - A Journal of Physical Sciences, Engineering and Technology (S-JPSET)*, vol. 14, no. 1, 2022, pp. 103-107. **[UGC Care Listed]**
- 8) Santosh Trimbak Warpe, Ravi Singh Pippal, "Newton Backward Difference application for automating fertilizer distribution in Agriculture using Wireless Sensor Network," *Agricultural Science Digest*, vol. 42, no. 2, 2022. **[UGC Care Listed]**
- 9) Rajesh Kumar Sharma, Ravi Singh Pippal, "Fault tolerant system design in Internet of Things network with blockchain validation," *Samriddhi - A Journal of Physical Sciences, Engineering and Technology (S-JPSET)*, vol. 13, no. 1, 2021, pp. 53-58. **[UGC Care Listed]**
- 10) Priyank Nayak, Ravi Singh Pippal, "Cryptanalysis of Zhian Zhu's Scheme and Evaluation of TMIS Smart Card Authentication Schemes," *Journal of Scientific Research*, vol. 13, no. 2, 2021, pp. 407-413. **[UGC Care Listed]**
- 11) Girdhar Gopal Ladha, Ravi Kumar Singh Pippal, "An efficient distance estimation and centroid selection based on k-means clustering for small and large dataset," *International Journal of Advanced Technology and Engineering Exploration*, vol. 7, no. 73, 2020, pp. 234-240. **[Indexed in SCOPUS]**
- 12) Puneet Dwivedi, Ravi Singh Pippal, "Analysis of DTN Routing Protocols by varying Simulation Run," *Journal of the Maharaja Sayajirao University of Baroda*, vol. 54, no. 2, 2020, pp. 151-156. **[UGC Care Listed]**
- 13) Rajesh Kumar Sharma, Ravi Singh Pippal, "Survey of Blockchain Validation Techniques in the Design of Secure IoT Network", *Journal of the Maharaja Sayajirao University of Baroda*, 2020, vol. 54, no. 2, pp. 144-150. **[UGC Care Listed]**
- 14) Rajesh Kumar Sharma, Ravi Singh Pippal, "IoT Gateway Security Mechanism for Efficient Network Establishment using Blockchain Implementation", *Wesleyan Journal of Research*, 2020, vol. 6, no. 5, pp. 2347-2354. **[UGC Care Listed]**
- 15) Santosh Trimbak Warpe, Ravi Singh Pippal, "Newton Backward Difference Application For Automating Fertilizer Distribution In Agriculture Using Wireless Sensor Networks," *GIS Science Journal*, vol. 7, no. 9, 2020, pp. 188-200.
- 16) Puneet Dwivedi, Ravi Singh Pippal, "Performance of Various DTN Routing Protocol under Several Mobility Models," *Journal of Critical Reviews*, vol. 7, no. 19, 2020, pp. 7087-7093.

- 17) Priyank Nayak, Ravi Singh Pippal, "A Robust Patient Authentication Scheme For Telecare Medicine Information Systems (TMIS) against Smart Card Security Breach," *Journal of Critical Reviews*, vol. 7, no. 19, 2020, pp. 4175-4183.
- 18) Puneet Dwivedi, Ravi Singh Pippal, "Performance Evaluation of Spray and Wait DTN Routing Protocol under Different Mobility Models Breach," *Journal of the Gujarat Research Society*, vol. 21, no. 16, 2019, pp. 186-192. **[UGC Care Listed]**
- 19) Girdhar Gopal Ladha, Ravi Singh Pippal, "An Efficient K-Means Method Based on Centroid Handling for the Similarity Estimation," *International Journal of Engineering and Advanced Technology (IJEAT)*, vol. 9, no. 2, 2019, pp. 4337-4341.
- 20) Ravi Singh Pippal, Nitika Singhi "Performance Analysis of LEACH-GAC over LEACH-B, LEACH-C and LEACH-GA," *Research Journal of Engineering Technology and Management*, vol. 1, no. 1, 2018, pp. 24-30.
- 21) Ravi Singh Pippal, Dharmendra Kumar "Performance Analysis of Multi-Level Hybrid Random Forest and Support Vector Machine based on K-Means for Feature Reduced Intrusion Detection," *Research Journal of Engineering Technology and Management*, vol. 1, no. 1, 2018, pp. 15-23.
- 22) Sandeep Dubey and Ravi Singh Pippal, "Secure, Selfish Replica Allocation Method for MANET", *Journal of Advanced Research in Dynamical & Control Systems*, vol. 10, no. 15, 2018, pp. 511-518.
- 23) Dharmendra Kumar and Ravi Singh Pippal, "Analysis of Data Normalization with Multilevel Classifiers for Intrusion Detection," *International Journal of Computational Engineering Research (IJCER)*, vol. 8, no. 9, 2018, pp. 61-67.
- 24) Dharmendra Kumar and Ravi Singh Pippal, "Performance Analysis of Multilevel Classifiers for Feature Reduced Intrusion Detection," *IOSR Journal of Engineering (IOSRJEN)*, vol. 8, no. 11, 2018, pp. 1-7.
- 25) Sandeep Dubey and Ravi Kumar Singh Pippal, "Efficient Selfish Node Management Method for MANET," *International Journal of Computer Sciences and Engineering (IJCSE)*, vol. 06, no. 10, 2018, pp. 334-338.
- 26) Nitika Singhi and Ravi Singh Pippal, "Impact Analysis of Attack in Multi-Hop LEACH Protocol in WSN," *International Journal of Computational Engineering Research (IJCER)*, vol. 8, no. 6, 2018, pp. 1-10.
- 27) Nitika Singhi and Ravi Singh Pippal, "Impact Analysis of Attack in Balanced LEACH Protocol in WSN," *IOSR Journal of Engineering (IOSRJEN)*, vol. 8, no. 6, 2018, pp. 26-36.
- 28) Pradeep Gupta, Ravi Singh Pippal, "Secure Framework for Administrator Authentication in UTM System," *National Journal of Multidisciplinary Research and Development (NJRMD)*, vol. 3, no. 1, 2018, pp. 453-457.
- 29) Sahil Khatri, Himanshu Saxena and Ravi Singh Pippal, "User Anonymity based Trusted Authentication Scheme for Wireless Environment," *National Journal of Multidisciplinary Research and Development (NJRMD)*, vol. 3, no. 1, 2018, pp. 458-462.
- 30) Tanmay Kasbe and Ravi Singh Pippal, "Enhancement in Diagnosis of Coronary Artery Disease using Fuzzy Expert System," *International Journal of Scientific Research in Computer Science, Engineering and Information Technology (IJSRCSEIT)*, vol. 3, no. 3, 2018, pp. 1324-1331.
- 31) Ravi Singh Pippal, "A Novel Smart Card Authentication Scheme using Invisible Image Watermarking," *SHODH-SANGAM-A RKDF University Journal of Science and Engineering*, vol. 01, no. 01, 2018, pp.1-10.
- 32) Ravi Singh Pippal, "Enhanced Advanced Smart Card based Password Authentication Scheme," *SHODH-SANGAM-A RKDF University Journal of Science and Engineering*, vol. 01, no. 01, 2018, pp.11-19.
- 33) Ravi Singh Pippal, "Identifying Weak Spots of Three Provably Secure Smart Card Authentication Schemes," *SHODH-SANGAM-A RKDF University Journal of Science and Engineering*, vol. 01, no. 01, 2018, pp.20-27.
- 34) Nitika Singhi, Ravi Singh Pippal, "Analysis of Key Management Schemes in MANET," *International Journal of Applied Environmental Sciences (IJAES)*, vol. 13, no. 02, 2018, pp. 161-169. **[To be indexed in SCOPUS]**
- 35) Sandeep Dubey, Ravi Singh Pippal, "Survey on Behavior of Selfish Node for Replica Allocation in MANET", *International Journal of Computer and Advanced Engineering Research (IJCAER)*, vol. 4, no. 4, 2017, pp. 18-23.

- 36) Sandeep Dubey, Ravi Singh Pippal, "A survey on Improved Data accessibility and data Updating Mechanism through Handling Selfishness in Replica Allocation over Mobile Ad-hoc Networks", *International Journal of Research in Technology (IJRT)*, vol. 4, no. 4, 2017, pp. 119-124.
- 37) Tanmay Kasbe and Ravi Singh Pippal, "Role of Fuzzy Expert System to Diagnosis Human Disease," *Journal of Emerging Technologies and Innovative Research (JETIR)*, vol. 3, no. 12, 2017, pp. 1-6.
- 38) Manish B. Giri, Ravi Singh Pippal, "A Pervasive System for Optimizing Water Utilization in Agriculture using Wireless Sensor Networks," *International Journal of Applied Environmental Sciences (IJAES)*, vol. 12, no. 08, 2017, pp. 1605-1615. **[To be indexed in SCOPUS]**
- 39) Santosh T Warpe and Ravi Singh Pippal, "A Study of Fertilizer Distribution System for Agriculture using Wireless Sensor Network," *International Journal of Computer Applications (IJCA)*, vol. 147, no. 2, 2016, pp. 43-46.
- 40) Manish B. Giri, Ravi Singh Pippal, "Agricultural Environmental Sensing Application Using Wireless Sensor Network for Automated Drip Irrigation," *International Journal of Computer Sciences and Engineering (IJCSE)*, vol. 04, no. 07, 2016, pp. 133-137. **(Cited By 01)**
- 41) Tanmay Kasbe and Ravi Singh Pippal, "Dengue Fever: State-of-the-Art Symptoms and Diagnosis," *International Journal of Computer Sciences and Engineering (IJCSE)*, vol. 04, no. 06, 2016, pp. 26-30. **(Cited By 01)**
- 42) Md. Jahid Akhtar, Rakesh Kumar Sanodiya and Ravi Singh Pippal, "Fuzzy Logic Controlled Resource Allocation for Efficient Load Balancing in Cloud Computing Environment," *International Journal of Engineering Sciences and Research Technology (IJESRT)*, vol. 5, no. 1, 2016, pp. 135-140.
- 43) Raksha Tomar, Murlidhar Vishwakarma and Ravi Singh Pippal, "Improving the Performance of Human Detection Technique using Cascaded Support Vector Machine," *International Journal of Computer Applications (IJCA)*, vol. 117, no. 3, 2015, pp. 1-4.
- 44) Monika Jhapate, Lalitesh Choudhary and Ravi Singh Pippal, "A Review on Triangle based Techniques in Biometric Gait Recognition," *International Journal of Computer Applications (IJCA)*, vol. 121, no. 7, 2015, pp. 6-9. **(Cited By 02)**
- 45) Smita Verma, Deepanjali Joshi and Ravi Singh Pippal, "Congestion Control Load Balancing Scheme Intended for Improving Performance of MANET," *International Journal of Engineering and Computer Science (IJECS)*, vol. 4, no. 6, 2015, pp. 12963-12969.
- 46) Roopesh Kumar Sharma, B. P. Patel, Y. K. Rana and R. S. Pippal, "Analysis of Routing Protocols for Data Aggregation in Wireless Sensor Network," *International Journal of Technology Research and Management (IJTRM)*, vol. 2, no. 2, 2015, pp. 1-6.
- 47) Nidhi Kaushal, Murlidhar Vishwakarma and Ravi Singh Pippal, "A Novel Set Level Technique for Image Segmentation Using Fuzzy Clustering and Self Organizing Map Network," *International Journal of Science and Research (IJSR)*, vol. 4, no. 6, 2015, pp. 830-833.
- 48) Thamraj N. Ghorsad, R. S. Pippal and Badri Prasad Patel, "Implementation of Hybrid Authentication System Based on Sound Signature and Graphics for Authentication," *Research Cell: International Journal of Engineering Sciences (IJES)*, vol. 14, no. 8, 2015, pp. 57-64.
- 49) Thamraj N. Ghorsad, R. S. Pippal and Badri Prasad Patel, "A Review on Schemes for User Authentication," *Research Cell: International Journal of Engineering Sciences (IJES)*, vol. 14, no. 7, 2015, pp. 47-56. **(Cited By 01)**
- 50) Shruti Khare, B. P. Patel, Ravi Singh Pippal and Y. K. Rana, "Review of lifetime optimization techniques in Wireless Sensor Networks," *International Journal of Technology Research and Management (IJTRM)*, vol. 2, no. 1, 2015, pp. 1-5.
- 51) Akhilesh Singh, Harsha Banafar and Ravi Singh Pippal, "Intrusion Detection on KDD99cup Dataset using K-means, PSO and GA: A Review," *International Journal of Electrical, Electronics and Computer Engineering (IJECE)*, vol. 4, no. 1, 2015, pp. 40-46.
- 52) Samiksha Chatare and Ravi Singh Pippal, "A Score and Cluster Head Based Algorithm for Finding MRC against Link Failures," *International Journal of Computer Science and Information Technologies (IJCSIT)*, vol. 6, no. 4, 2015, pp. 3619-3624.

- 53) Nidhi Ghodki, Deepanjali Joshi and Ravi Singh Pippal, "Detection and Prevention Scheme against Jamming Attack in MANET: A Survey," *International Journal of Scientific Progress and Research (IJSPPR)*, vol. 06, no. 01, 2014, pp. 37-41.
- 54) Smita Verma, Deepanjali Joshi and Ravi Singh Pippal, "Congestion Control Multipath Load Balancing Techniques in Mobile Ad hoc Network: A Survey," *International Journal of Scientific Progress and Research (IJSPPR)*, vol. 06, no. 02, 2014, pp. 75-80. **(Cited By 03)**
- 55) Ravi Singh Pippal, Jaidhar C. D. and Shashikala Tapaswi, "Robust Smart Card Authentication Scheme for Multi-server Architecture," *Wireless Personal Communications (Springer)*, vol. 72, no. 1, 2013. pp. 729-745. **[Indexed in SpringerLink, DBLP, ACM, SCOPUS] (Cited By 52)**
- 56) Ravi Singh Pippal, Jaidhar C. D. and Shashikala Tapaswi, "Enhanced Time-Bound Ticket-Based Mutual Authentication Scheme for Cloud Computing," *Informatica*, vol. 37, no. 2, 2013, pp. 149-156. **[Indexed in SpringerLink, DBLP, ACM, SCOPUS] (Cited By 12)**
- 57) Ravi Singh Pippal, Pradeep Gupta and Rakesh Singh, "A Novel Smart Card Authentication Scheme using Image Encryption," *International Journal of Computer Applications (IJCA)*, vol. 72, no. 9, 2013, pp. 8-14. **(Cited By 06)**
- 58) Ravi Singh Pippal, Pradeep Gupta and Rakesh Singh, "Dynamic Encryption Key Based Smart Card Authentication Scheme," *International Journal of Computer Applications (IJCA)*, vol. 72, no. 9, 2013, pp. 15-18. **(Cited By 04)**
- 59) Ravi Singh Pippal, Rajesh Ahirwar, Shivpratap Singh Kushwah and Pradeep Yadav, "A Secure SCAM (Smart Card based Authentication Mechanism)," *International Journal of Computer Applications (IJCA)*, vol. 72, no. 5, 2013, pp. 26-31. **(Cited By 03)**
- 60) Ravi Singh Pippal, Shashikala Tapaswi and Jaidhar C. D., "Secure Key Exchange Scheme for IPTV Broadcasting," *Informatica*, vol. 36, no. 1, 2012, pp. 47-52. **[Indexed in SpringerLink, DBLP, ACM, SCOPUS] (Cited By 13)**
- 61) Ravi Singh Pippal, Jaidhar C. D. and Shashikala Tapaswi, "Security Issues in Smart Card Authentication Scheme," *International Journal of Computer Theory and Engineering (IJCTE)*, vol. 4, no. 2, 2012, pp. 206-211. **(Cited By 05)**

Conference Publications (36)

- 1) Vinod Krishna Sethi, Sudesh Kumar Sohani, Ravi Kumar Singh Pippal, Meenakshi Samartha, "Strategies of Installation of a Solar Integrated Carbon Capture and Sequestration (CCS) Plant on a 500 MW Size Coal fired Thermal Power Plant in India," Accepted for publication in 4th *International Conference on Trends and Recent Advances in Civil Engineering (TRACE – 2022)*, 18-19 October 2022, Noida, India. **[Will be indexed in SCOPUS]**
- 2) Meenakshi Samartha, Ravi Kumar Singh Pippal, Vinod Krishna Sethi, "Biofuel Production through Algae Route using Solar Assisted Carbon Capture Plant," Accepted for publication in *IOP Conference Series: Earth and Environmental Science*, 1084, 2022. **[Will be indexed in SCOPUS]**
- 3) Ravi Singh Pippal, Vinod Krishna Sethi, "Chemistry of Carbon Dioxide for Recycling," Accepted for publication in *AIP Conference Proceedings*, 2022. **[Will be indexed in SCOPUS]**
- 4) Priyank Nayak, Ravi Singh Pippal, "Session Migration Scheme among Cipher Machines using Polynomials," In: *Proceedings of the International Conference on Material Science and Manufacturing Engineering (ICMSME-2021)*, December 10-11, 2021.
- 5) Rajesh Sharma, Ravi Singh Pippal, "Malicious Attack and Intrusion Prevention in IoT Network Using Blockchain Based Security Analysis," In: *Proceedings of the 12th International Conference on Computational Intelligence and Communication Networks (CICN-2020)*, 2020, pp. 380-385. **[Indexed in IEEE Xplore, SCOPUS]**
- 6) Puneet Dwivedi, Ravi Singh Pippal, "Design and Performance Analysis of Routing Protocols for Delay Tolerant Networks," Accepted for oral presentation in 5th *IEEE International Conference on Recent Advances and Innovations in Engineering (ICRAIE 2020)*, 1-3 December 2020, Jaipur, India. **[Indexed in IEEE Xplore, SCOPUS]**
- 7) Santosh T. Warpe, Ravi Singh Pippal, "Automated Intensification of Fertilizer Distribution in Agriculture Using Wireless Sensor Network," A Study of Precision Agriculture Techniques for Fertilizer Distribution",

In: Proceedings of the 1st International Conference on Innovative Trends and Advances in Engineering and Technology (ICITAET-2019), December 27-28, 2019, Buldhana, Maharashtra, India. [Indexed in IEEE Xplore, SCOPUS]

- 8) Preeti Verma, Ravi Singh Pippal, Sunil Patil, "A Machine Learning Approach for Solar Radiation Assessment using Multispectral Satellite Images," *In: Proceedings of the 3rd International Conference on Recent Advances in Smart Innovative Ideas With Multidisciplinary Research (IC-RASIMR-2019), November 30 - December 01, 2019, Bhopal, India.*
- 9) Santosh T. Warpe, Ravi Singh Pippal, "A Study of Precision Agriculture Techniques for Fertilizer Distribution", *In: Proceedings of the 2nd International Conference on Contemporary Technological Solutions towards the fulfillment of Social Needs, September 28-29, 2019, Bhopal, India.*
- 10) Paritosh Goldar, Ravi Singh Pippal, "A Survey of Routing Optimization Techniques for Device-to-Device Communications in Internet of Things", *In: Proceedings of the 2nd International Conference on Contemporary Technological Solutions towards the fulfillment of Social Needs, September 28-29, 2019, Bhopal, India.*
- 11) Girdhar Gopal Ladha, Ravi Kumar Singh Pippal, "A computation analysis to predict diabetes based on data mining: A review," *In: Proceedings of the International Conference on Communication and Electronics Systems, 2018, Coimbatore, India, pp. 6-10. [Indexed in IEEE Xplore, SCOPUS]*
- 12) Gagan Sharma, Ravi Singh Pippal, "Methodological Survey on Big Data in Business Process Evolution and Management," *In: Proceedings of the Conference on Contemporary Technological Solutions towards fulfillment of Social Needs, August 31, 2018, Bhopal, India, pp. 155-157.*
- 13) Nitika Singhi, Ravi Singh Pippal, "Exploit Valuation of Energy-useful Secure-Genetic Algorithm in Wireless Sensor Network" *In: Proceedings of the 16th IRF International Conference, 2018, New Delhi, India, pp. 53-60.*
- 14) Nitika Singhi, Ravi Singh Pippal, "An Energy Efficient Secure GAC Algorithm in Wireless Sensor Network," *In: Proceedings of the 4th International Conference on VLSI, Communication and Networks (VCAN-2018), 2018, Alwar, India, pp. 271-278.*
- 15) Puneet Dwivedi, Ravi Singh Pippal, "Introduction to Delay Tolerant Network and its Associated Routing Protocols," *In: Proceedings of the National Conference on New Era of Technological Interconnection in Electrical, Electronics and Computer Engineering, September 14-15, 2018, Bhopal, India, pp. 15.*
- 16) Puneet Dwivedi, Ravi Singh Pippal, "Study of routing protocols for delay tolerant networks," *In: Proceedings of the Conference on Contemporary Technological Solutions towards fulfillment of Social Needs, August 31, 2018, Bhopal, India, pp. 144-147.*
- 17) Sandeep Dubey, Ravi Singh Pippal, "Efficient replica allocation in ad hoc networks for developed data accessibility," *In: Proceedings of the Conference on Contemporary Technological Solutions towards fulfillment of Social Needs, August 31, 2018, Bhopal, India, pp. 120-124.*
- 18) Rajesh Sharma, Ravi Singh Pippal, "Survey of Cloud Virtualization Management and Placement Methods," *In: Proceedings of the Conference on Contemporary Technological Solutions towards fulfillment of Social Needs, August 31, 2018, Bhopal, India, pp. 158-161.*
- 19) Manish Kumar Sahu, Ravi Singh Pippal, "Review of malware analysis, classification and detection techniques," *In: Proceedings of the Conference on Contemporary Technological Solutions towards fulfillment of Social Needs, August 31, 2018, Bhopal, India, pp. 152-154.*
- 20) Jitendra Patel, Ravi Singh Pippal, "Big data analytics for the detection of structure based attack over social media: A review," *In: Proceedings of the Conference on Contemporary Technological Solutions towards fulfillment of Social Needs, August 31, 2018, Bhopal, India, pp. 13-16.*
- 21) Rishank Rathore, Ravi Singh Pippal, "Social theories based unsupervised community detection over social media: A review," *In: Proceedings of the Conference on Contemporary Technological Solutions towards fulfillment of Social Needs, August 31, 2018, Bhopal, India, pp. 100-105.*
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- 23) Dharmendra Kumar, Ravi Singh Pippal, "A comprehensive review on intrusion detection system and techniques," In: *Proceedings of the Conference on Contemporary Technological Solutions towards fulfillment of Social Needs*, August 31, 2018, Bhopal, India, pp. 133-137.
- 24) Manish B. Giri, Ravi Singh Pippal, "Use of Linear Interpolation for automated Drip Irrigation system in Agriculture using Wireless Sensor Network," In: *Proceedings of the International Conference on Energy, Communication, Data Analytics and Soft Computing (ICECDS-2017)*, August 01-02, 2017, Poonamallee, Tamil Nadu, India. **[Indexed in IEEE Xplore, SCOPUS]**
- 25) Tanmay Kasbe, Ravi Singh Pippal, "Design of Heart Disease Diagnosis System using Fuzzy Logic," In: *Proceedings of the International Conference on Energy, Communication, Data Analytics and Soft Computing (ICECDS-2017)*, August 01-02, 2017, Poonamallee, Tamil Nadu, India. **[Indexed in IEEE Xplore, SCOPUS]**
- 26) Girdhar Gopal Ladha, Ravi Kumar Singh Pippal, "A review and analysis on data mining methods to predict diabetes," In: *Proceedings of the 7th International Conference on Communication Systems and Network Technologies*, 2017, Bhopal, India, pp. 334-337. **[Indexed in IEEE Xplore, SCOPUS]**
- 27) Dharmendra Kumar, Ravi Singh Pippal, "Risk Assessment and Management of Network Security Incidents through Organization Culture and its Challenges," In: *Proceedings of the IEEEFORUM International Conference*, August 20, 2017, New Delhi, India, pp. 67-73.
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- 30) Ravi Singh Pippal, Jaidhar C. D. and Shashikala Tapaswi, "A Novel Smart Card Mutual Authentication Scheme for Session Transfer among Registered Devices," In: *Proceedings of the 3rd IEEE International Advance Computing Conference (IACC-2013)*, February 22-23, 2013, Ghaziabad, India, pp. 1540-1547. **[Indexed in IEEE Xplore, SCOPUS] (Cited By 04)**
- 31) Ravi Singh Pippal, Jaidhar C. D. and Shashikala Tapaswi, "Highly Secured Remote User Authentication Scheme using Smart Cards," In: *Proceedings of the 7th IEEE International Conference on Industrial Electronics and Applications (ICIEA-2012)*, July 18-20, 2012, Singapore, pp. 1001-1005. **[Indexed in IEEE Xplore, SCOPUS] (Cited By 06)**
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- 34) Ravi Kumar Singh, Deepti Jain and Shraddha Singh, "Implementing Different Variants of RSA and Comparing them based on Hardware Fault Attack," In: *Proceedings of the 1st International Conference on Advances in Computing (ICAC-2008)*, February 21-22, 2008, Chikhli, India.
- 35) Sanjay Silakari and Ravi Kumar Singh Pippal, "An Exhaustive Analysis of Current Routing Protocols in MANETs," In: *Proceedings of the 10th International Symposium on Wireless Personal Multimedia Communications (WPMC-2007)*, December 3-6, 2007, Jaipur, India, pp. 740-43.
- 36) Sanjay Silakari and Ravi Kumar Singh Pippal, "Prevention from Selfish Nodes in MANETs through extension of AODV Protocol," In: *Proceedings of the 2nd International Conference on Advanced Computing and Communication Technologies (ICACCT-2007)*, November 3, 2007, Panipat, India.

Short Term Training Programs/ Workshops/ Seminars Attended (16)

- 1) Participated in the two-day National Level Workshop on “**NIRF INDIA RANKINGS - 2022 For Higher Educational Institutions**” organized by Institute for Academic Excellence in collaboration with Collegiate Education & Technical Education Department, Govt. of Telangana on January 05-06, 2022.
- 2) Participated in the one-day Workshop on “**OER and How to create MOOC Courses/ Utility of MOOC Courses for Regular/ Open Universities**” organized by Raj Bhawan, Bhopal on February 28, 2020.
- 3) Participated in the one-day Workshop on “**Criteria-VII (INSTITUTIONAL VALUES AND BEST PRACTICES)**” organized by Barkatullah University, Bhopal on October 24, 2019.
- 4) Participated in the one-day Workshop on “**Criteria-V (STUDENT SUPPORT AND PROGRESSION)**” organized by Barkatullah University, Bhopal on October 12, 2019.
- 5) Participated in the one-day Workshop on “**Criteria-IV (INFRASTRUCTURE AND LEARNING RESOURCES)**” organized by Barkatullah University, Bhopal on October 05, 2019.
- 6) Participated in the one-day Workshop on “**Workshop on Criteria-I**” organized by Barkatullah University, Bhopal on September 19, 2019.
- 7) Participated in the one-day Workshop on “**Assessment of Higher Education Institutions in Madhya Pradesh by National Assessment and Accreditation Council (NAAC)**” organized by Barkatullah University, Bhopal on September 15, 2019.
- 8) Participated in the Knowledge Sharing Program on “**Entrepreneurship and new Venture creation (the current scenario & trends in India, a live case study of an entrepreneurship venture)**” organized by RKDF University, Bhopal on April 20, 2019.
- 9) Participated in the 4th National Conference on “**Skilling in Higher Education**” organized by BSNL, Bhopal on March 27, 2019.
- 10) Participated in the one-day Workshop on “**National Academic Depository (NAD)**” organized by University Grants Commission, CRO, Bhopal at BSSS College, Bhopal on July 25, 2018.
- 11) Participated in the one-day workshop on “**Virtual Labs**” organized by IIT Delhi and SRK University, Bhopal held on August 27, 2016.
- 12) Participated in the FDP (Faculty Development Programme) on “**Artificial Intelligence & Soft Computing Techniques**” conducted by Deptt. of CSE, UIT at RGPV, Bhopal held on April 28-May 02, 2015.
- 13) Participated in the FDP (Faculty Development Programme) on “**Cloud Computing, Information Security**” conducted by TCS (Tata Consultancy Services) at ABV-Indian Institute of Information Technology and Management, Gwalior held on May 11, 2011.
- 14) Participated in two-days Training Programme on “**LaTeX: The Language of Scientific Writing (LSW 2010)**” at ABV-Indian Institute of Information Technology and Management, Gwalior held on September 4-5, 2010.
- 15) Participated in one-day Workshop on “**How to Get Published in Internationally Refereed Journals: A Practical Guide**” organized by ABV-Indian Institute of Information Technology and Management, Gwalior held on March 21, 2010.
- 16) Participated in two-days Workshop on “**Security Risks in Internet (SRI-09)**” organized by ABV-Indian Institute of Information Technology and Management, Gwalior held on December 18-19, 2009.

Paper Presentation International / National Conferences (10)

- 1) “Session Migration Scheme among Cipher Machines using Polynomials,” *In: Proceedings of the International Conference on Material Science and Manufacturing Engineering (ICMSME-2021)*, December 10-11, 2021.
- 2) “Design and Performance Analysis of Routing Protocols for Delay Tolerant Networks,” Accepted for oral presentation in *5th IEEE International Conference on Recent Advances and Innovations in Engineering (ICRAIE 2020)*, 1-3 December 2020, Jaipur, India. [Indexed in IEEE Xplore, SCOPUS]

- 3) "A computation analysis to predict diabetes based on data mining: A review," *In: Proceedings of the International Conference on Communication and Electronics Systems*, 2018, Coimbatore, India, pp. 6-10. **[Indexed in IEEE Xplore, SCOPUS]**
- 4) "Enhancing MSF for Mobile ad hoc Network Security through active Handshaking and Multipath," *In: Proceedings of the International Conference on Emerging Trends in Science, Engineering & Management (ETSEM-2018)*, February 24-25, 2018, Alwar, India.
- 5) "Comments on Biometrics Based Remote User Authentication Scheme using Smart Cards," *In: Proceedings of the National Conference on Wired and Wireless Networks and Security Issues (WWNSI-2013)*, April 26-27, 2013, Bhopal, India, pp. 38-41.
- 6) "A Novel Smart Card Mutual Authentication Scheme for Session Transfer among Registered Devices," *In: Proceedings of the 3rd IEEE International Advance Computing Conference (IACC-2013)*, February 22-23, 2013, Ghaziabad, India.
- 7) "Comparative Analysis of Smart Card Authentication Schemes," *In: Proceedings of the 4th IEEE International Conference on Advanced Computing and Communication Technologies (ICACCT-2010)*, October 30, 2010, Panipat, India, pp. 113-18.
- 8) "Implementing Different Variants of RSA and Comparing them based on Hardware Fault Attack," *In: Proceedings of the 1st International Conference on Advances in Computing (ICAC-2008)*, February 21-22, 2008, Chikhli, India.
- 9) "An Exhaustive Analysis of Current Routing Protocols in MANETs," *In: Proceedings of the 10th International Symposium on Wireless Personal Multimedia Communications (WPMC-2007)*, December 3-6, 2007, Jaipur, India, pp. 740-43.
- 10) "Prevention from Selfish Nodes in MANETs through extension of AODV Protocol," *In: Proceedings of the 2nd International Conference on Advanced Computing and Communication Technologies (ICACCT-2007)*, November 3, 2007, Panipat, India.